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Forecasting potential bark beetle outbreaks based on spruce forest vitality using hyperspectral remote-sensing techniques at different scales

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ABSTRACT

The bark beetle (*Ips typographus L.*) is known for the detrimental impact it can have on Europe's mature spruce forests with bark beetle outbreaks already having devastated thousands of hectares of spruce forests in Germany. This study analysed the hypothesis that the vitality of spruce vegetation is already susceptible from factors such as climate change or emissions to a certain extent before infestation, so that the role of the subsequent bark beetle infestation is only secondary.

Hyperspectral remote-sensing techniques were used to detect changes in biochemical–biophysical vegetation characteristics in the spruce forest of the Bavarian Forest National Park, Germany. For this study, several spectral bands, vegetation indices and specific spectral band combinations of hyperspectral HyMAP remote-sensing data with a 4 m and a 7 m ground resolution were analysed and compared in terms of their classification accuracy, generating an ID3 decision tree.

The vitality classes and thus also the attack stages of the spruce vegetation could be estimated with moderate to good accuracy using hyperspectral remote-sensing data. Clear spectral differences between the class with spruce trees that were still green but with reduced vitality (possibly the first stages of green-attack) and the class with healthy spruce trees could be ascertained. The best spectral characteristics, spectral indicators and spectral derivatives related to vitality classes and thus attack stages were typically based on wavebands related to prominent chlorophyll absorption features in the VI within the spectral range of 450–890 nm. Only limited spectral information and derivatives could be found in the short-wave infrared region 1 (SWIR) within the spectral range of 1400–1800 nm, which reflects the water content of the spruce needles. The class of spruce trees that were still green but with reduced vitality (possibly the first stages of green-attack) showed a trend towards detectability and differentiation with spectral indicators and index derivatives. However, the prediction of observed effects with 64% accuracy as observed here is regarded as insufficient in forestry practises. Hyperspectral data with a ground resolution of 4 m were found to contain more information relevant to estimating the vitality class of spruce vegetation compared to hyperspectral data with a ground resolution of 7 m.

1. Introduction

In Europe and North America, forests are experiencing some of the worst outbreaks of insects in history. It therefore comes as little surprise that bark beetles (Scolytinae) are often considered to be one of the most significant pests in forestry. Not only can bark beetle outbreaks have devastating impacts on the lumber industry, but they are regarded by some to be one of the most significant natural disturbances to spruce forests (Schelhaas et al., 2003; Raffa et al., 2008). For this reason, Slobodyan and Slobodyan, 2001 also refer to the European spruce bark beetle (*Ips typographus L.)* as a biological indicator of disturbances in the functioning of forest ecosystems.

In Europe 154 species of bark beetle are known, whereby each species is adapted to only one or a few host tree species. The European (or eight-spined) spruce bark beetle *Ips typographus* is known to infest conifers – predominantly spruces (*Picea abies L.*). In the European spring time (March/April) the adult bark beetles start to swarm from daytime temperatures around 16.5°C.

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Subsequently, they attack spruces (*Picea abies L.*) and lay their eggs beneath the bark, where the larvae hatch and start to eat away at the inner bark of the infested trees. This feeding behaviour attacks the living phloem tissues of the trees, meaning that the transport of nutrients from the leaves to the roots is increasingly impaired. Due to this secondary starvation of nutrients in the leaves, there is a gradual reduction in chlorophyll and ultimately chlorophyll decay (Marx, 2010). From the middle of June through to the end of July the imagoes then leave their host tree and attack new spruces in their second generation. Trees that have only recently been infested are still green (green-attack), with the reddish-brown colouring of the spruce crowns (red-attack) only occurring after the beetles have already left the host tree that was used for breeding.

The devastating impacts of bark beetle on working forests have prompted research to improve our understanding of the different behaviour patterns and outbreak behaviour of bark beetles. For example Rossi et al. (2009), Avtzis et al. (2012) and Six (2012) have investigated the biology, development, ecology and phylogeography of different bark beetle species. From a different approach, the spatial distribution and pattern of *I. typographus* populations in Europe has been analysed by Heurich et al., 2003, Meier et al. (2003), Müller et al. (2008), Ogris and Jurc (2010) and Lausch et al. (2011, 2013). In addition, a number of other investigations have been conducted for North America (Powers et al., 1999; Negron et al., 2000; Klutsch et al. 2009) and Canada (Wulder et al., 2006a; Aukema et al., 2008; Coops et al., 2009). Due to this imminent threat, there has also been a recent research focus on monitoring bark beetles, in particular the efficiency of different monitoring techniques and strategies of bark beetle (Jurc et al., 2006; Meurisse et al., 2008; Saeed et al., 2010). Various systems have already been tested to monitor bark beetle infestations such as pheromone beetle traps (Liu and Dai, 2006), trap trees (Zumr and Stary, 1993) and spruce stand transects walked by foresters (Marx, 2010). Consequently, important information can be collected about the current infestation situation, its progression as well as swarming behaviour (Saeed et al., 2010). However, the disadvantage of such monitoring techniques is that they are not only time-consuming but also costly and thus unsuitable for forest stands that are larger than 2000 ha (Marx, 2010). Standardised methods therefore need to be established that enable a timely and cost-effective monitoring of the presence, distribution and outbreak potential of the bark beetle in spruce forests.

Remote-sensing techniques could provide the answer, as they have already been used to directly map and assess different issues of bark beetle and spruce vegetation behaviour characteristics. Numerous studies using remote-sensing techniques have concentrated particularly on the recognition and detection of the earliest bark beetle infestation of spruce. Depending on the length of the bark beetle attacks, one is able to differentiate between the different stages of bark beetle attack of the spruce vegetation. Trees that have only been infested very recently are still green (green-attack), while the reddish-brown colouring of the spruce crowns (red-attack) usually occurs when the beetles have already left the host tree used for breeding. It is only 36 months after the bark beetle attack that the spruces start to appear grey in colour (grey-attack). The symptoms of changes in biochemical–physical parameters of spruce vegetation with red and grey attack has been proven using remote-sensing techniques (Franklin et al., 2003; Skakun et al. 2003; Wulder et al., 2006a). However, so far research on "green attack" detection using remote-sensing sensors has been inconclusive, with forest managers still relying on high spatial resolution data of red-attack for planning ground surveys to establish green-attack and for implementing mitigation actions for infested areas.

Bark beetle outbreak pattern analysis and assessment based on MODIS time-series data (Eklundh et al., 2009), Landsat TM (Thematic Mapper) data (Coops et al. 2006, 2009; Wulder et al. 2006b; Hais and Kučera, 2008) and SPOT data has also been conducted by Wulder et al. (2006a). Furthermore, high spatial resolution data such as GeoEye-1 (Dennison et al., 2010), Ikonos (White et al. 2005), RapidEye (Marx, 2010) and Quickbird multispectral image (Hicke and Logan, 2009), Hyperspectral Mapper data (HyMAP, Hatalla et al., 2010; Fassnacht et al. 2012) have been used to detect and assess canopy mortality during mountain pine beetle outbreaks.

It can therefore safely be said that remote-sensing techniques have so far been used successfully to monitor spruce stands already infested with bark beetle. Currently however we are not aware of any available method that uses remote-sensing techniques to assess or predict potential areas of bark beetle infestation in the future. It is well-known that healthy forests are carbon sinks, but forests that are killed by bark beetles release carbon, turning them into significant carbon sources. An anticipation of where potential outbreaks might occur could assist policy-makers with climate change mitigation, by preserving forest carbon stocks as a result of implementing measures aimed at preventing bark beetle attacks in those areas where outbreaks can be predicted from hyperspectral remote-sensing.

It could be feasible to record or predict potential areas of bark beetle infestation when the following hypothesis can be applied and quantified using remote-sensing techniques.

Local capture experiments with the bark beetle have shown that the infestation of spruces by no means occurs systematically (Fahse and Heurich, 2011), but are rather subject to infestation criteria, on which there is still insufficient knowledge.

- According to Fahse and Heurich (2011) when there is an outbreak of bark beetle in a region, the predisposition of the spruce vegetation does not play a role in bark beetle infestation in a subsequent year. When the beetles invade, it is rather the case that the beetles "randomly" infest all trees.
- With a low level of bark beetle infestation in a region, however the predisposition of the spruce trees does play a large role in bark beetle infestation in a subsequent year. At times that are not during an outbreak there are not enough beetles available to trigger off an infestation of healthy trees. The beetles are thus restricted to engaging in a reproductive strategy that involves the "*selective infestation*" of trees that are already weak (Fahse and Heurich, 2011).

In the case of a "selective infestation of spruces" Fahse and Heurich (2011) suppose that the spruces have already been so damaged in terms of their vitality due to various factors i.e. climate change, a systematic increase in temperature, a reduction in the ground water level and thus the supply of water, as well as an increase in ozone and particle emissions, that the subsequent bark beetle infestation should only be regarded as a secondary event i.e. the bark beetle can be regarded as a secondary pest.

Progressive climate change as well as high levels of air pollution can lead to severe changes in the context of the bark beetle. Progressive warming as well as a lack of water over longer periods of time reduces the "health" or "vitality" of spruce forests. This makes the spruces more susceptible to an attack from *Ips typographus* (Ohrn, 2012). The after-effects are evident. Insects such as *I. typographus* rely on the temperature for their metabolism and population development. An increase in the mean temperature can result in an earlier onset of the flight of spruce bark beetle in the spring. Moreover, an increase in the mean temperature means that the eggs hatch sooner i.e. they develop into adults sooner, leading to a faster development of a second generation (Jönsson et al., 2007, 2009; Faccoli, 2009; Ohrn, 2012). Furthermore, high elevation spruce forest ecosystems like the Bavarian National Park in

Germany are highly sensitive to a high level of air pollution (including SO_2, NO_x, O_3). In particular, the location of the Bavarian National Park, which is located directly on the border to the Czech Republic – one of the most heavily polluted regions in the world (Ardö et al. 1997; Campbell et al., 2004; Hickler et al., 2012) has dramatically altered the vitality status of spruce vegetation.

Spectral and geometrical high resolution remote-sensing techniques are now being implemented more and more to characterize the ''health'' or ''vitality'' of plants and vegetation. To quantify and assess the health or ''vitality'' of plants and vegetation, a combination of different indicators have been used. The most important functional reactions of stress or the vitality of vegetation in the context of remote-sensing techniques are changes in biochemical and biophysical characteristics of forest vegetation. The most important of these are summed up in the following. (1) Stress in vegetation leads to changes in the composition and proportion of photosynthetically-active pigments such as chlorophyll a and b, xanthophyll and vegetation greenness (Miller et al. 2002; Gitelson et al., 2003; Zhang et al., 2008; Verrelst et al., 2010). (2) Stress in vegetation leads to a modification of photosynthetic activity through the fluorescence effects in plants (Naumann et al., 2008; Zarco-Tejada et al., 2009). (3) Vegetation stress changes the water status of the plant as well as the percentage of water in leaf cells (Eitel et al., 2006; Suárez et al. 2008; Zarco-Tejada et al., 2012). (4) Vegetation stress can also cause changes to plant transpiration, resulting in changes to leaf arrangement and geometry, stomata distribution as well as special protective mechanisms such as leaf hairs and cuticula in its structure, and ultimately the loss of needles and leaves (Claudio et el., 2006; Polak et al., 2006; Hernández-Clemente et al., 2011) as well as (5) changes to and modification of cellulose and lignin content in the vegetation (Bannari et al., 2006; Swatantran et al., 2011).

High spectral and geometrical remote-sensing data therefore provide the opportunity to analyse the predisposition of spruce vegetation to bark beetle infestations in terms of its vitality.

The main objectives of this study were therefore to test and evaluate the suitability of hyperspectral remote-sensing techniques in detecting the predisposition of spruce vegetation to bark beetle damage. To accomplish this, spectral indicators and index derivatives were quantified to differentiate between different levels of vitality of spruce tree stands to find the most suitable spectral information for the model prediction of potential bark beetle outbreaks in spruce trees. Furthermore, we aimed to investigate the predictive power of the model on potential bark beetle outbreaks from hyperspectral remote-sensing techniques on different spatial scales.

2. Data and methods

2.1. Study area

To model the prediction of bark beetle outbreaks we used the study area Rachel Lusen Region – also referred to as the ''old region'' of the Bavarian Forest National Park (Lat. E.13.23, Long. N.48.53). This region covers approximately 130 km^2 of the Bavarian National Park (Fig. 1).

The Bavarian Forest National Park was established in 1970. The first outbreaks of *Ips typographus (L.)* occurred in 1992 (initially at higher elevations) with outbreaks peaking around 1996 and 1997 but continuing to this day. From 1988 to 2010 a total of 5800 ha of the natural Norway spruce vegetation died off as a result of *Ips typographus* infestations (Fig. 1, Lausch et al. 2011).

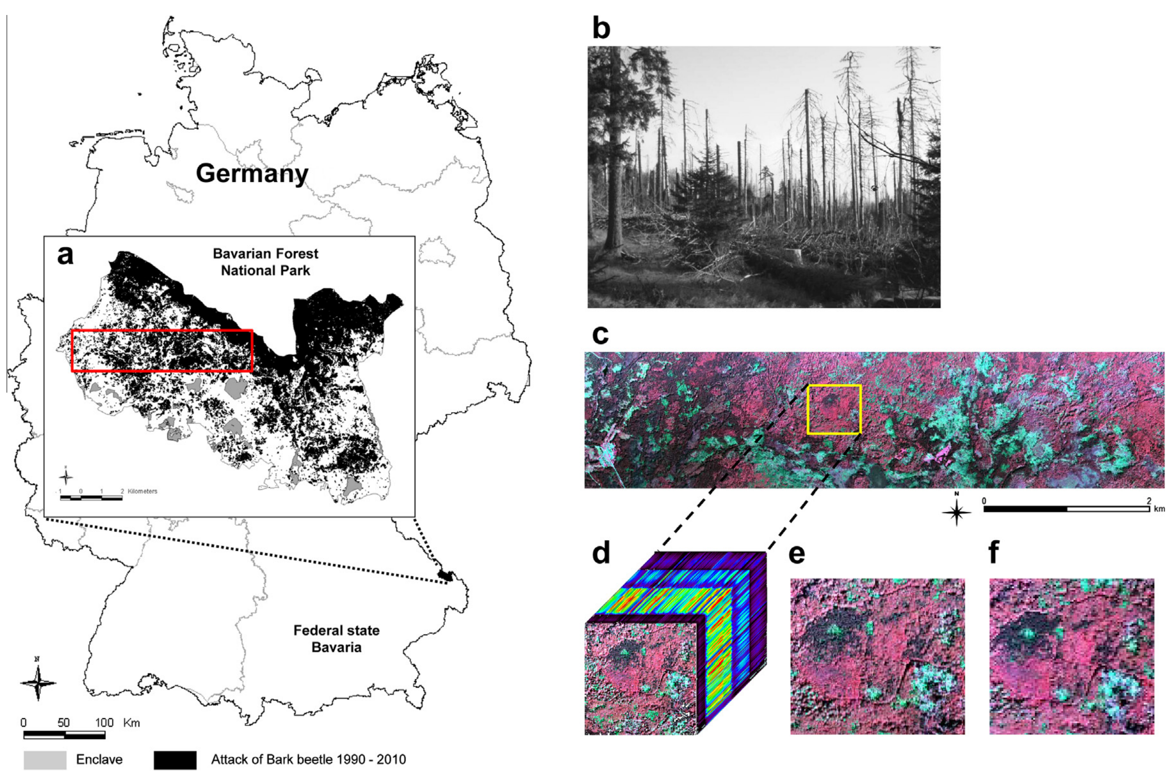

Fig. 1. (a) Location of the study area – The Rachel Lusen Area of the Bavarian Forest National Park ''old part'', Germany, in black – Bark beetle outbreaks in the Rachel Lusen Area (''old part'') of the Bavarian Forest National Park, Germany (1990–2010), (b) Bark beetle outbreak in 2003 in Bavarian Forest National Park (c) Colour Infrared (CIR) image – Bavarian Forest National Park region, taken from the hyperspectral sensor HyMAP, spectral range 450–2480 nm, 125 spectral bands, date of recording 2010-09-25, ground resolution of 4 m, (d) Data cube with 125 spectral bands, (e) ground resolution of 4 m and (f) ground resolution of 7 m.

2.2. Geo-data

2.2.1. Reference data of bark beetle outbreaks from 1988 to 2011

To monitor and control the impact of these bark beetle outbreaks, from 1988 to 2011 aerial image campaigns were conducted every year within the National Park area. A detailed report is available that was produced by the Bavarian Forest National Park administration (2002) about the methodology that was implemented for taking, processing and interpreting these aerial images. Flight campaigns took place whereby 267 CIR aerial images were taken during every flight with a geometrical resolution of 20 cm, equating to a total of 147 gigabytes of data. Every year there were two CIR flight campaigns with the first one always taking place in June/July. Spruces suffering from a bark beetle infestation can only be recognised from August onwards, thus campaigns carried out during the months of June/July still allow any additional dead wood that has occurred since the previous year to be recorded. In September/October of every year, a second campaign was then carried out. These aerial images can then be used to assess how many dead trees have accumulated in the year of investigation. Colour infrared stereo aerial images were taken at scales between 1:10,000 and 1:15,000. The images were mapped using a stereoscope and transparent slides, before being drawn onto maps and finally digitalised up until the year 2000 (Heurich et al., 2010). As technology became more sophisticated, aerial images were then scanned with a photogrammetric scanner at a resolution of between 15 and 20 μm (from the year 2001). A block triangulation and an ortho-image calculation could then be conducted based on these scans. ERDAS StereoAnalyst was then used to make an assessment of dead wood patches, as it enables a 3D-view of forest stands. StereoAnalyst was used to overlap the results of analyses from the previous year with current aerial images, allowing any changes from the year before to be directly delineated on the screen stereoscopically (Heurich et al., 2010). Areas with less than 5 trees were not mapped.

2.2.2. Airborne hyperspectral data

2.2.2.1. Data specifications and pre-processing. Flight campaigns were run by the German Aerospace Centre (DLR) in August 2009 using the Australian commercial airborne hyperspectral sensor HyMAP (Hyperspectral Mapper) with a ground resolution of 4 m and 7 m – date of recording 2009-08-25 (Fig. 1). The sensor has a wavelength interval of 450–2480 nm with 125 spectral bands (Cocks et al., 1998). Other parameters of the recorded imaging hyperspectral data can be found in Table 1.

Geometric and atmospheric correction of the images was conducted by the DLR based on the parametric correction software ATCOR4 (Richter and Schläpfer, 2002) and ORTHO. Only level two data were delivered by the data provider. Due to their geometry and structure, forest stands are strongly subjected to the effects of anisotropy and canopy shadow (Schlerf et al., 2005). Therefore, in order to correct the HyMAP data for illumination effects, a cross-track illumination correction was subsequently performed using ENVI 4.7 Imaging Processing Software (ITT Visual Information Solutions, 2009).

2.2.2.2. Calculation of spectral vegetation indices and spectral derivatives. In the current study several hyperspectral indices and spectral derivatives were calculated based on the HyMAP data and tested according to their suitability to differentiate between spruce classes based on their vegetation stress level.

The spectral indices used were categorised into different spectral derivatives: (I) *Reflectance VI* – we used the reflectance value ($R_{(X)}$) at the central wavelength (x nm) of each band of the imaging HyMAP spectrometer (450–2480 nm, 125 spectral bands), (II) *Vegetation Indices* – vegetation indices were used from the relevant literature (Table 2) and (III) $VI_{(xy)}$ *spectral band combinations* – we tested combinations of the 125 HyMAP bands and used a formula that also includes two bands. The formulas tested are a subtraction, addition, multiplication and division of the spectral bands.

2.3. Definition of spruce classes

The current vector information for bark beetle infestation (cf. Section 2.1) is fundamental to mask out the hyperspectral HyMAP data. Five vitality classes for spruce vegetation were determined that form the basis of the modelling (Table 3, Fig. 2).

Class 1 (C1 – dry1, grey–green) includes all spruce stands infested by bark beetle between 1988 and 2008. The spruces on these sites have died off, been felled or already removed from the forest. In these areas the forest has already started to regenerate. Class 2 (C2 – dry 2, red–grey) represents those stands that have died off due to a bark beetle outbreak in 2009. Here the needles and spruces are a reddish-brown to grey colour with or without needle loss and are dying off or have already died off. For class 3 (C3, infestation 2010) bark beetle outbreak vector data from 2010 were used. In this class spruce needles are still green but display reduced vitality – stage 1. The hyperspectral HyMAP data were recorded on 25.08.2009. At this point in time the masked infestation areas of 2010 had not been infested with *Ips typographus*. The infestation

Table 1
Specifications of the hyperspectral data HyMAP (Hyperspectral Mapper) for the investigation.

Recording date	Pixel size (m)	Flight altitude (m)	Flight direction	FOV (°)	Swath (km)	Spectral range (nm)	Spatial pixel	Spectral resolution (nm)		Spectral bands	Signal to noise ratio	Platform
2009-08-25	4	2929	W–E	61,3	2.0	450–2480	512	VIS	450–890 = 15–16	125	<500:1	Aircraft
								NIR	890–1350 = 15–16			DO 228-212
								SWIR1	1400–1800 = 15–16			
								SWIR2	1950–2480 = 18–20			
2009-08-25	7	3200	N–S	61,3	3.6	450–2480	512	VIS	450–890 = 15–16	125	<500:1	Aircraft
								NIR	890–1350 = 15–16			DO 228-212
								SWIR1	1400–1800 = 15–16			
								SWIR2	1950–2480 = 18–20			

Table 2
Spectral vegetation indices (VI) that were calculated as known to have a relationship with vegetation stress.

Index	Short	Algorithm	Reference
Leaf pigments			
Anthocyanin reflectance index 2	ARI 2	R800 * [(1/R550)–(1/R700)]	Gitelson et al. (2002)
Carotinoid reflectance index 1	CRI 1	[(1/R510)–(1/R550)]	Gitelson et al. (2002)
Greenness			
Normalized difference vegetation index	NDVI	(R800 – R670)/(R800 + R670)	Rouse et al. (1974) and Lichtenthaler et al. (1996)
Modified normalized difference index	MND	(NIR – (1.2 * red)–)/NIR + red)	Tucker (1979)
Green normalized difference vegetation index	GNDVI	(R780 – R550)/(R780 + R550)	Gitelson et al. (1996) and Babar et al. (2006)
Ratio	DI1	(R800–R550)	Buschman and Nagel (1993)
Pigment activity/light use efficiency			
Structure insensitive pigment index	SIPI	(R800 – R445)/(R800 + R680)	Penuelas et al. (1994)
Normalized pigment chlorophyll ratio index	NPCI	(R680 – R430)/(R680 + R430)	Penuelas et al. (1994)
Red edge index			
Vogelmann 2	Vog 2	(R734 – R747)/(R715 + R726)	Vogelmann et al. (1993) and Zarco-Tejada and Miller (1999)
Ratio index	Br625_795	(R625/R795)	Carter and Miller (1994) and Pontius et al. (2008)
Water indices			
Normalized difference soil moisture index	NSMI	[(R1800 – R2119)/(R1800 + R2119)]	Haubrock et al. (2008)
Moisture stress index	MSI	(R1599/R819)	Hunt and Rock (1989) and Ceccato et al. (2001)
Soil and atmosphere adjusted			
Red edge position	REP	700 + 40 [(((R670 + R780)/2)–R700)/(R740–R700)]	Cho and Skidmore (2006)
Transformed chlorophyll absorption in reflectance index	TCARI	3 * [(R700–R670) – 0.2 * (R700–R550)(R700/R670)]	Haboudane et al. (2002)
Green optimized soil adjusted vegetation index	GOSAVI	(1 + 0.16)(R800–R670)/(R800 + R670 + 0.16)	Rondeaux et al. (1996)

R_x stands for reflectance at wavelength x nm.

Table 3
Determination of spruce vitality classes based on different levels of vegetation stress.

Class	Attribute	Characteristics of spruce vegetation	Bark beetle outbreak data[a]
C1	Dry 1(grey–green)	Areas with forest loss showing signs of forest regeneration	1988–2008[a]
C2	Dry 2 (red–grey)	Needles and spruces are a reddish-brown to grey colour with or without needle loss. They are dying off or have already died off Grey and red-attack of bark beetle	2009[a]
C3	Infestation 2010	Spruce needles are green with reduced vitality Bark beetle attack is still not visible (possible green-attack) Bark beetle infestation of these sites occurred in 2010	2010[a]
C4	Infestation 2011	Spruce needles are green showing vitality Bark beetle attack is still not visible Bark beetle infestation of these sites occurred in 2011	2011[a]
C5	Healthy (green)	Spruce needles are green showing vitality No bark beetle infestation of these sites from 1988 to 2011	2012

[a] Data sets provided by the Bavarian Forest National Park, Germany.

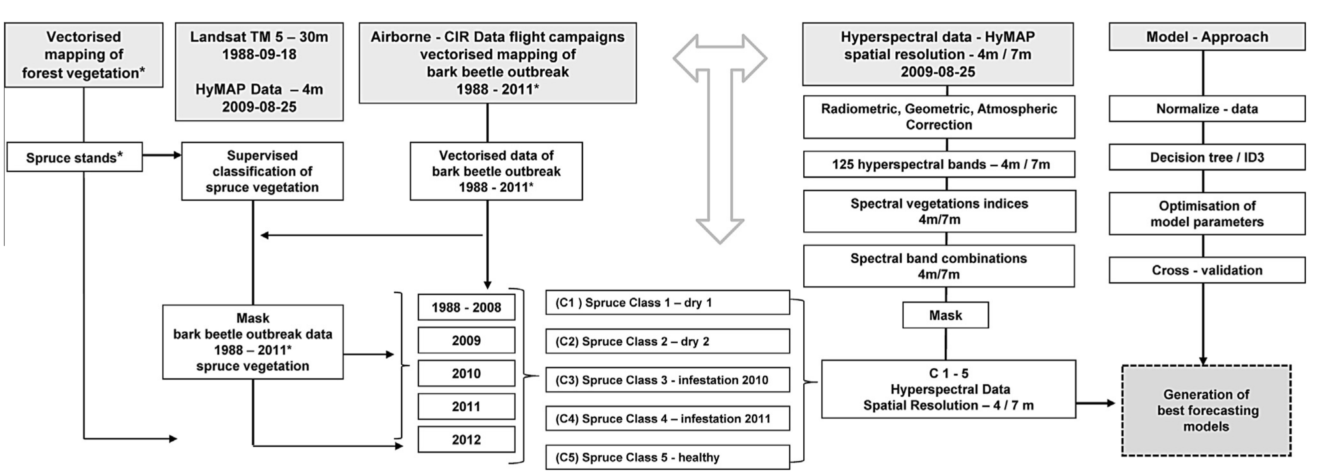

Fig. 2. Flow chart of data selection and model approach to evaluate the forecasting of potential bark beetle outbreaks in the Bavarian Forest National Park, Germany.

of these sites occurred in 2010. For class 4 (infestation 2011) spruce needles are still green and showing vitality. The spruces have still not been attacked by *Ips typographus*, with a bark beetle infestation of these sites occurring in 2011. For class 5 (C5, healthy, green) spruce needles are green and showing vitality. There are still no signs of vegetation stress or bark beetle infestation. These spruces were not infested with bark beetle until 2011 (Table 3).

2.4. Model approach

The model approach to analysing indicators of forecasting potential bark beetle outbreaks based on hyperspectral remote-sensing data is shown in Fig. 2. Existing vector data on bark beetle outbreaks from 1988 to 2011 (cf. Section 2.2) was used to mask out the spruce classes C1 to C4 for the hyperspectral HyMAP data (spatial resolution of 4 m and 7 m, recording date 2009-08-25). To determine the spruce class C5 (healthy), a spruce vegetation classification was carried out using the Landsat TM5 (recording date 1988-09-18) and HyMAP data (recording date 2009-25-08) together with the additional use of existing vector information on spruce stands in the Bavarian Forest National Park. Spruce stands that had previously been susceptible to bark beetle infestations from 1988–2011 were then discounted from the remaining spruce stands determined in the HyMAP classification, resulting in the spruce vitality class 5 (healthy) in Fig. 2. For each spruce vitality class the following spectral information was ascertained: (i) spectral wavelength, (ii) spectral vegetation indices, as well as (iii) spectral derivatives by means of calculating different band combinations.

The explorative analysis that was conducted using all data showed that the spectral characteristics investigated by sampling spruce classes were neither normally distributed nor did they show any similar variances. As a feasible approach to differentiate between spruce classes we therefore turned to methods of data-mining decision tree algorithms. Since the 1990s different decision tree algorithms have been successfully implemented for solving classification issues with remote-sensing data (Lawrence and Wright, 2001; Mewes et al., 2008). For the differentiation and the prediction function of the five spruce vitality classes we used the ID3 (Iterative Dichotomiser 3) decision tree classification method based on Quinlan (1993). The ID3 algorithm proceeds iteratively. Here, for every attribute that is not used, entropy calculations are performed with respect to the amount of training. The attribute containing the most information and thus the smallest entropy is selected and a new knot is generated from it. The procedure ends when all training instances have been classified, i.e. when a classification has been assigned to every knot until one reaches a sheet (Whitten et al., 2011). To validate the performance of all models, a ten-fold cross-validation method with stratified sampling was applied (Efron and Tibshirani, 1993). As performance criteria for the predictive power of the spruce vitality classes, both the statistical indicators classification accuracy and the Kappa index (Cohen, 1968) were used.

The calculation of all spectral vegetation indices of imaging hyperspectral data was carried out using ENVI/ IDL, v.4.7. The statistical and data-mining analyses were conducted using ENVI/IDL v. 4.7., RapidMiner, v.5.2 (Rapid-I GmbH, Dortmund, Germany).

3. Results and discussion

This paper examined the hypothesis whether the vitality of spruce vegetation is already impaired by factors such as climate change or emissions to a considerable extent before infestation, so that the subsequent bark beetle infestation only plays a secondary role in reducing vitality. Using high-resolution airborne hyperspectral remote-sensing techniques combined with digitalised data of bark beetle outbreaks from 1988 to 2011 we investigated whether differences in biochemical–biophysical vegetation characteristics can be differentiated according to different levels of vegetation stress.

3.1. Data and model approach

To investigate the different spectral patterns of the five spruce classes as a function of spruce tree vitality, the spruce classes were masked out according to the method described in Section 2.4. To test the five spruce classes, the areas that were masked out were overlayed with aerial imagery data from 2004 and 2010. Here we discovered that in particular the border regions of the vectorised infested areas from two consecutive years of infestation had been erroneously assigned. The five spruce classes that had been derived were thus once again subjected to an extensive manual correction process by overlaying them with aerial image mosaics from 2004 to 2010.

To find the best model for forecasting potential bark beetle outbreaks based on the vitality status of spruce forest vegetation, the classification algorithms Naive Bayes, Support Vector Machine – SVM, Library for Support Vector Machines – LibSVM after Chang and Lin (2002) and Decision Tree (ID3) were compared in terms of their classification accuracy (Gordalla, 2012). A 10-fold cross-validation procedure was used to assess the predictive power of the three different classification algorithms. Out of the three classification algorithms, the SVM model demonstrated the best classification performance. However, for the classification algorithm SVM, the set of spectral indicators included in the model was far too extensive, making a concrete designation of specific indicators impossible. Although the decision tree classification achieved a somewhat lower overall accuracy performance of the spruce classes, by using the ID3 (Iterative Dichotomiser 3) decision tree classification method based on Quinlan (1993), the structure of the decision tree is kept relatively small and an overfitting of the model is avoided (Witten et al., 2011). Furthermore, numerous studies have already confirmed that the decision tree classification algorithm has great potential for classification approaches using hyperspectral remote-sensing techniques (Xu et al., 2006; Mewes et al., 2008; Bakos and Gamba, 2011; Monteiro and Murphy, 2011). Consequently, the ID3 decision tree classification method was used to classify the vitality status of HyMAP spectral data in this study.

3.2. Indicators to determine different vitality stages of spruce vegetation

Fig. 3 shows the spatial characteristics of the five different vitality classes of spruce vegetation using HyMAP hyperspectral remote-sensing data with a ground resolution of 4 m and 7 m as well as data from RGB aerial imagery from 2010.

The mean spectral characteristics for the classes analysed (dry 1, dry 2, infestation 2010, infestation 2011 and healthy) can be differentiated by their reflectance potential for the wavelengths 450–2480 nm (Fig. 4a and b). The spectral reflectance for dry 1 and dry 2 is considerably higher in the range from 450 to 700 nm than for the classes infestation 2010 (C3) and infestation 2011 (C4) as well as class 5 – healthy (C5). A similar pattern of the mean spectral response curves for the five spruce vitality classes could also be observed in the spectral range of 1500–1900 nm and 2100–2480 (Fig. 4a and b).

On the basis of the decision tree (ID3) we analysed the predictive power of spectral information, spectral derivatives and spectral indices to differentiate between the five spruce classes: class 1 – dry 1 (grey–reen), class 2 – dry 2 (red–grey), class 3 – infestation

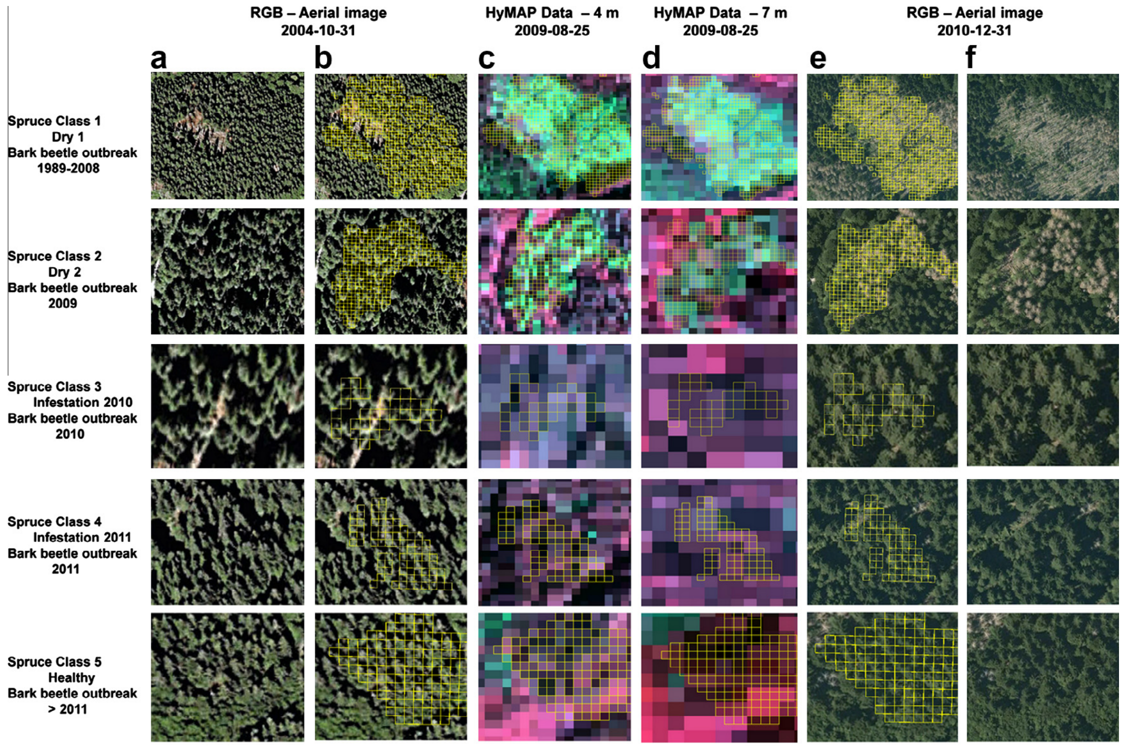

Fig. 3. Figure showing the five different spruce vegetation classes with their five different stress levels, (a and b) RGB aerial image from 2004, (c) Hyperspectral HyMAP data with 4 m ground resolution, shown as colour infrared (CIR) (d) Hyperspectral HyMAP data with 7 m ground resolution, shown as colour infrared (CIR) and (e and f) RGB aerial image from 2010.

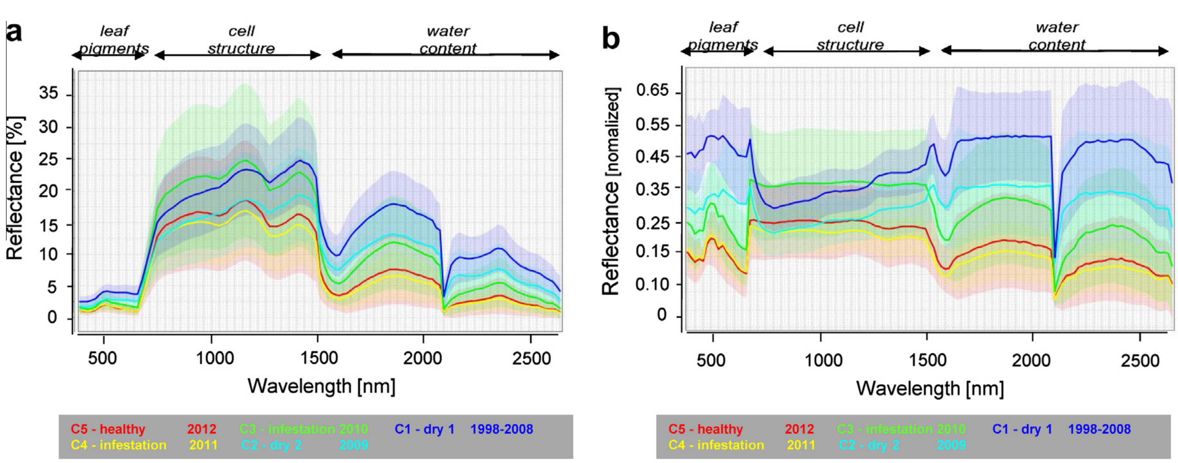

Fig. 4. (a) Mean spectral reflectances of HyMAP data (spatial resolution of 4 m) for the five spruce vitality classes: class 1 – dry 1 (grey–green), class 2 – dry 2 (red–grey), class 3 – infestation 2010, class 4 – infestation 2011, class 5 – healthy and (b) For a better comparison of images, the mean spectral reflectances of the HyMAP data for the spruce classes were normalized between 0 and 1.

2010, class 4 – infestation 2011 and class 5 – healthy (green). The best model fit for classification using hyperspectral HyMAP data was achieved with a total classification accuracy of 51.56% with a Kappa value of 0.395. Here the highest class recall was ascertained with 71.25% for class 1 (dry 1) and the lowest class recall was ascertained with 28.12% for class 3 (infestation 2010) (Table 4).

Class 1 covers all spruce stands infested by the bark beetle from 1988 to 2008. The spruces on these sites had either died off, been felled or already removed from the forest. In these areas the forest had already started to regenerate. Early stages of forest regeneration leads to a change in the spectral signal resulting from an increase in the photosynthetic pigments and water content as well as changes in the cell structures of young spruce vegetation. Consequently, a clear designation of the dead wood spectral is difficult. For the differentiation of class 5 (C5, healthy – green) a classification accuracy of 61.56% was achieved for the model. This class shows no sign of vegetation stress or infestation with bark beetle. For all other spruce classes investigated only a very low classification accuracy of 28–48% could be achieved. When one looks at the spectral assignment for generating the model with the relevant spectral indicators and index derivatives, then it is evident that the most significant indicators for differentiating between the

Table 4
Classification matrix of the best run to distinguish between the spruce classes: class 1 – dry 1 (grey–green), class 2 – dry 2 (red–grey), class 3 – infestation 2010, class 4 – infestation 2011, class 5 – healthy (green) using the DT (ID3) classifier and their best spectral predictors for this run using HyMAP hyperspectral data with a ground resolution of 4 m, $n = 320$.

Spruce classes	True dry 1	True dry 2	True infestation 2010	True infestation 2011	True healthy	Class precision (%)	Total classification accuracy	Kappa
Pred. dry 1	228	99	30	9	12	60.32		
Pred. dry 2	71	154	30	14	13	54.61		
Pred. infestation 2010	7	19	90	18	44	50.56		
Pred. infestation 2011	10	26	75	156	54	48.60		
Pred. healthy	4	22	95	123	197	44.67		
Class recall	71.25%	48.12%	28.12%	48.75%	61.56%		51.56%	0.395%
Best spectral predictors for this model:								
Wavelengths (nm)	Spectral derivatives (nm)							
455	[(R690/R718)]							
528	[(R528/R646)]							
690	[(R718/R1948)]							
	[(R455 + R528)]							
	[(R690 – R1404)]							

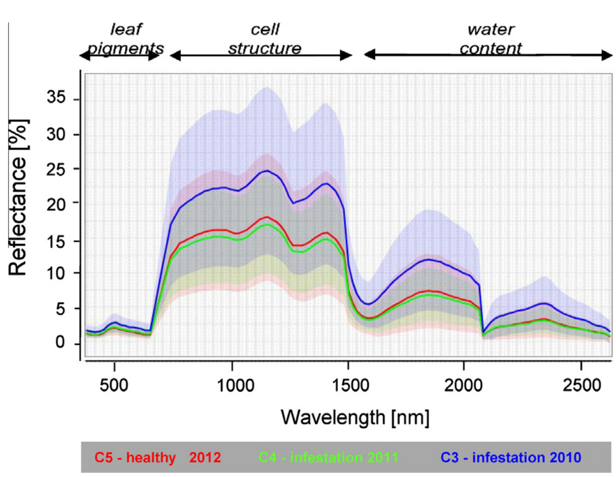

Fig. 5. Mean spectral reflectances of HyMAP data (spatial resolution of 4 m) for the three spruce classes: class 3 – infestation 2010, class 4 – infestation 2011, class 5 – healthy.

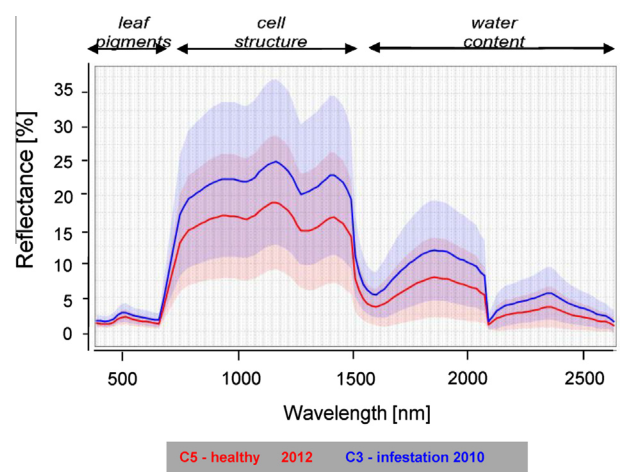

Fig. 6. Mean spectral reflectances of HyMAP data (spatial resolution of 4 m) for the five spruce vitality classes: class 3 – infestation 2010 and class 5 – healthy.

spruce vegetation vitality classes C1, 2, 3, 4 and C5 are predominantly concentrated in the VI (Visible Infrared) and the NIR (Near Infrared) range over a spectral range of 400–800 nm (Fig. 8, Table 9).

Another analysis investigated the extent to which the spruce vegetation that had still not been attacked in 2009 (C3, bark beetle infestation took place in 2010) differs in its spectral characteristics from spruce class 4 – infestation 2011 as well as class 5 – healthy (C5). The mean spectral reflectance (Fig. 5) of the three different spruce classes already shows a strong spectral similarity between

Table 5
Classification matrix of the best run for differentiating between spruce classes: class 3 – infestation 2010, class 4 – not attacked 2, class 5 – healthy (green) using the DT (ID3) classifier and their best spectral predictors for this run using HyMAP hyperspectral data with a ground resolution of 4 m, $n = 320$.

Spruce classes	True infestation 2010	True infestation 2011	True healthy	Class precision (%)	Total classification accuracy	Kappa
Pred. infestation 2010	150	38	65	59.29		
Pred. infestation 2011	104	203	119	47.65		
Pred. healthy	66	79	136	48.40		
Class recall	46.88%	63.44%	42.50%		50.94%	0.264%
Best spectral predictors for this model:						
Wavelengths (nm)	Spectral derivatives	Spectral indices (VI)				
455	[(R455–R690)]	ARI 2				
	[(R469/R528)]	NSMI				
	[(R528/R1404)]	REP				
	[(R646–R1404)]	[(R710–760)]				
	[(R646/R2065)]					
	[(R690–R1404)]					

Table 6
Classification matrix of the best run to distinguish between the spruce classes: class 3 – infestation 2010 and class 5 – healthy (green) using the DT (ID3) classifier and their best spectral predictors for this run using HyMAP hyperspectral data with a ground resolution of 4 m, $n = 320$.

Spruce classes	True infestation 2010	True healthy	Class precision (%)	Total classification accuracy	Kappa
Pred. infestation 2010	237	147	61.72		
Pred. healthy	83	173	67.58		
Class recall	74.06%	54.06%		64.06%	0.281%
Best spectral predictors for this model:					
Spectral indices (VI)					
ARI 2	GOSAVI				
CRI 1	NDVI				
DI 1	NPCI				
GNDVI	NSMI				
MSI	TCARI				

Table 7
Classification matrix of the best run to distinguish between the spruce classes: class 4 – infestation 2011and class 5 – healthy (green) using the DT (ID3) classifier and their best spectral predictors for this run using HyMAP hyperspectral data with a ground resolution of 4 m, $n = 320$.

Spruce classes	True infestation 2011	Infestation 2011	Class precision (%)	Total classification accuracy	Kappa
infestation 2010	192	113	62.96		
infestation 2011	128	207	61.79		
Class recall	60.00%	64.69%		62.34%	0.247%
Best spectral indicators for differentiation					
Wavelengths (nm)	Spectral derivatives	Spectral indices (VI)			
455	[(R484 + R1388)]	ARI 2			
1404	[(R646 + R2478)]	GOSAVI			
2331	[(R1388 + R2478)]	MND 705			
		R605_760			
		[(R710–760)]			
		VOG 2			

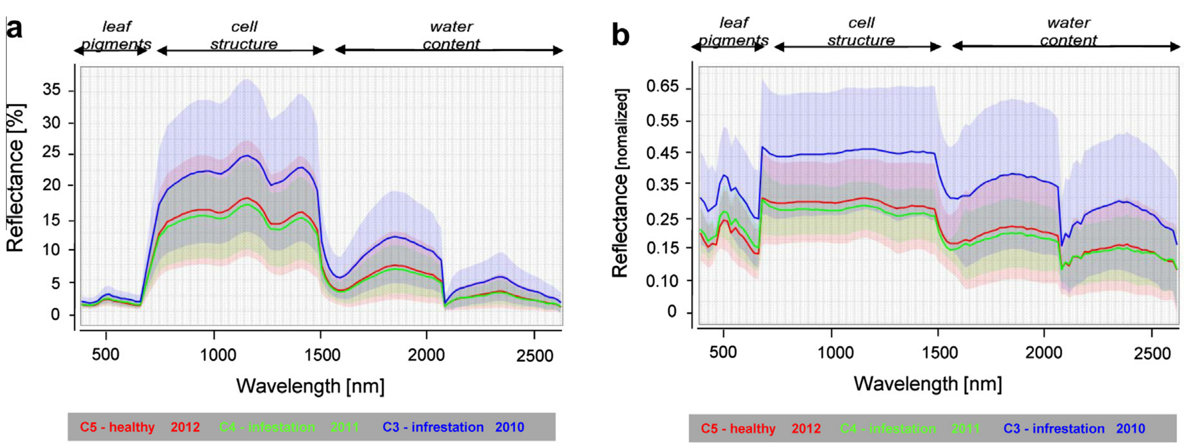

Fig. 7. (a) Mean spectral reflectances of HyMAP data (spatial resolution of 4 m) for the five spruce classes: class 1 – dry1 (grey–green), class 3 – infestation 2010 and class 5 – healthy and (b) Mean spectral reflectances of HyMAP data for the spruce classes were normalized between 0 and 1 for better image comparison.

class 4 (infestation 2011) and class 5 (healthy). Between both of these spruce vitality classes there is an overlay of spectral signatures.

The poor differentiation that is already visible in the mean spectral reflectance curves of the HyMAP data, can also be seen in the predictive model. Here the best model fit was obtained to differentiate between the spruce classes 3, 4 and 5 with a total classification accuracy of 50.89% (Kappa = 0.264) (Table 5). The mean spectral reflectance value shows that there is a considerable overlap in the spectral signatures of spruce class 4 (infestation 2011) and class 5 (healthy) (Fig. 5).

Another important question that we wanted to answer from the current study was whether or not the vitality of the spruce stands had already been made susceptible by other factors before infestation, so that for this tree species in particular the bark beetle can only be regarded as a secondary pest. To answer this question, the spectral characteristics of spruce vitality class 3 (still green but with reduced vitality and although not infested in 2009 possibly showing the first stages of green-attack) were compared with those of spruce vitality class 5 (healthy). Based only on the mean spectral response curves of both classes, a differentiation can already be made between both classes (Fig. 6).

The best model fit for the decision tree classification between spruce classes 3 (infestation 2010) and 5 (healthy) was achieved with a total classification accuracy of 64.06% (Table 6). Here, spruce class 3 that had not been attacked was differentiated with an accuracy of 74.06%, whereas for class 5, a class recall of 54.06% was achieved. The most significant spectral indicators and index derivatives for the decision tree classification of spruce class 3 (infestation 2010) against spruce class 5 (healthy) are in this respect: MSI

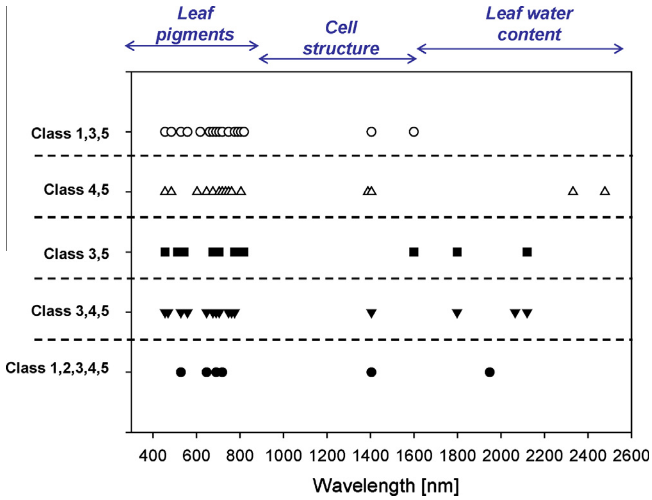

Fig. 8. Location of the best spectral predictors (spectral indices, spectral derivatives, spectral wavelengths) with their specific wavelengths between 450 and 2480 nm for the best model fits based on decision tree classification (ID3) for spruce classes 1–5, class 1 – dry 1 (grey–green), class 2 – dry 2 (red–grey), class 3 – infestation 2010, class 4 – infestation 2011, class 5 – healthy (green).

(Moisture Stress Index); CRI 1 (Carotinoid Reflectance Index 1), GNDVI (Green Normalized Difference Vegetation Index), ARI 2 (Anthocyanin Reflectance Index 1), NWI 2 (Normalized Water Index 2), NDVI (Normalized Difference Vegetation Index), NSMI (Normalized Difference Soil Moisture Index), GOSAVI (Green Optimized Soil Adjusted Vegetation Index), NPCI (Normalized Pigment Chlorophyll ratio Index), TCARI (Transformed Chlorophyll Absorption Reflectance Index) and DI1 (Derivative Index) respectively.

An important test at this stage was to compare the spectral characteristics of class 4 – infestation 2011 (class 4, bark beetle infestation occurred in 2011) against spruce class 5, i.e. all healthy and non-infested spruce vegetation. The classification matrix of the best run for this analysis only shows a low total classification accuracy of 62.34% (Table 7).

In another analysis, only the separability of spruce class 1 (dry 1) compared to class 3 (infestation 2010) and class 5 (healthy) with spectral indicators and index derivatives was tested for. Here, the mean spectral reflectance curves of HyMAP data with a spatial resolution of 4 m showed a good differentiation of these three spruce classes (Fig. 7).

Based on decision tree classification (ID3), the best prediction of the class 1 (dry 1), class 3 (infestation 2010) and class 5 (healthy) was achieved with a total classification accuracy of 69.27% (Kappa = 0.539). In this case, the best recall was achieved for spruce class 1 with 89.06%, for the recall class 3 with 60.31% and for class 5 with 58.44% (Table 8). The best spectral indicators and spectral derivatives as predictors for class-differentiation between class 1 (dry 1), class 3 (infestation 2010) and class 5 (healthy) can be found in Table 8.

Fig. 8 summarizes the location of the spectral wavelength characteristics from the best spectral indicators, spectral derivatives and individual wavelengths in the spectral range between 450 and 2480 nm of the HyMAP data for the best model fits based on decision tree classification (ID3) for all spruce classes analysed.

The results in Fig. 8 show that the best spectral indicators are particularly concentrated in the wavelength region between 450 and 900 nm. In these regions it is biochemical changes in particular and characteristics of the leaf pigments that can be depicted. Only a few spectral indicators are concentrated in the wavelength region between 1400 and 2480 nm, where changes in leaf water content between the classes of spruce vegetation play a significant role.

To differentiate between spruce vegetation vitality, in particular the spectral indicators and spectral derivatives in the range of the spectral wavelength range from 450 to 800 nm were used in the decision tree (Table 9). These results also confirm studies undertaken by Campbell et al. (2004), Heath (2001) and Wulder et al. (2006a). Campbell et al. (2004) used hyperspectral data in a similar way to detect the initial stages of forest damage at the canopy level of Norway spruce (*Picea abies L.)*. They were able to

Table 8
Classification matrix of the best run to distinguish between the spruce classes: class 1 – dry 1 (grey–green), class 3 – infestation 2010, class 5 – healthy (green) using the (ID3) decision tree classifier and their best spectral predictors for this run with HyMAP hyperspectral data – ground resolution 4 m, $n = 320$.

Spruce classes	True dry 1	True infestation 2010	True healthy	Class precision (%)	Total classification accuracy	Kappa
Pred. dry 1	285	23	16	87.96		
Pred. infestation 2010	28	193	117	57.10		
Pred. healthy	7	104	187	62.75		
Class recall	89.06%	60.31%	58.44%		69.27%	0.539
Best spectral indicators for differentiation						
Spectral derivatives	Spectral indices (VI)					
[(R528/R660)]	MSI					
[(R528/R718)]	Br 625_795					
[(R455/R1404)]	REP					
[(R1404/R1948)]	SIPI					
[(R528/R660)]	DI1					
	NPCI					

Table 9
Classification matrix of the best run fits to distinguish among the five spruce classes: class 1 – dry 1 (brown), class 2 – dry 2, class 3 – infestation 2010, class 4 – infestation 2011, class 5 – healthy (green) based on Published VI, Reflectance VI, VI formula combinations and all three VI types for HyMAP hyperspectral data – ground resolution 7 m.

Spruce classes	True 1	True 2	True 3	True 4	True 5	Class precision (%)	Total classification accuracy	Kappa
Pred. 1	51	48	33	24	34	26.84		
Pred. 2	41	55	33	29	37	28.21		
Pred. 3	37	33	56	33	44	27.59		
Pred. 4	31	23	43	80	36	37.56		
Pred. 5	34	35	29	28	43	25.44		
Class recall	26.29%	28.35%	28.87%	41.24%	22.16%		29.38%	0.117

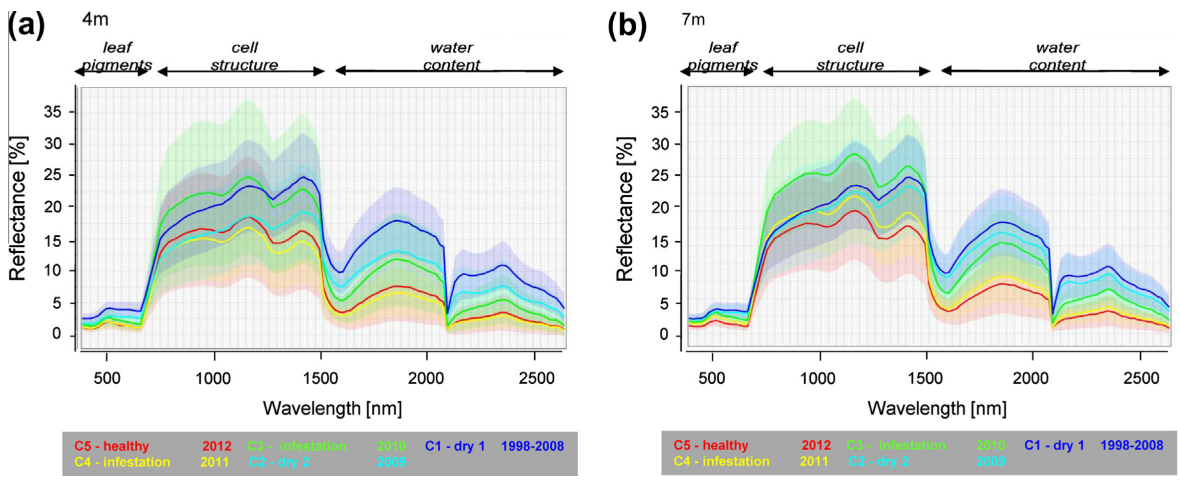

Fig. 9. Mean spectral reflectances of HyMAP data with a spatial resolution of (a) 4 m and (b) 7 m for the five spruce classes: class 1 – dry1 (grey–green), class 2 – dry2 (red–grey), class 3 – infestation 2010, class 4 – infestation 2011, class 5 – healthy.

separate the different stages of vitality within the 673–724 nm hyperspectral waveband regions. In our study we also used spectral indicators and index derivatives in the 1500–2500 nm hyperspectral waveband region to differentiate between the stages of spruce vitality. This substantiates the results from Nebeker et al. (1993) and Wulder et al. (2006a). They identified spruce stress compared to non-visual symptoms of stress by detecting changes in the leaf water level of spruce vegetation. Changes in leaf water content are sensitive in the spectral range from 1300 to 2500 nm (Wulder et al, 2006a; Borzuchowski and Schulz, 2010; Cheng et al. (2011). Ahern (1988) used the hyperspectral sensor CASI (Compact Airborne Spectrographic Imager) to detect differences in the spectral reflectance response of attacked trees compared to non-attacked trees. These differences manifest in an increase in the spectral reflectance of red wavelengths and a decrease in the green reflectance (Ahern, 1988; Curran et al., 1990). Ahern (1988) on the other hand conducted these investigations at the leaf level, but cautioned an extrapolation to the entire spruce vegetation as understory vegetation and branches etc. could dilute the signal. A comparison of the spectral reflectances of spruce vegetation that had been attacked or not attacked was carried out by Heath (2001) on the vegetation level using hyperspectral remote-sensing data, demonstrating changes in the spectral range from 750 to 900 nm. He also indicated however that there is a strong overlap between the spectral reflectances of 'green-attacḱ trees and healthy trees. A strong overlap and thus also the difficulty of separating spruce vegetation that has not been attacked (C3, C4) and healthy spruce vegetation (C5) was also confirmed by our investigations. According to Wulder et al. (2006a) the difficulty of detecting bark beetle damage of spruce vegetation with remote-sensing techniques lies in the detection of fade rates, whereby foliage changes its colour as a result of phenology or vegetation stress such as climate change or air pollution. All of the investigations that have been conducted by Wulder et al. (2006a) so far to differentiate between the spectral signatures of healthy and newly infested trees, trees that are still green and those without the first signs of the red–brown change in crown colour, have always been unsuccessful because there is a strong overlap in the spectral signatures of healthy and newly infested trees. According to Wulder et al. (2006a), the mapping, detection and differentiation of bark beetle outbreak attack stages and the differentiation between 'green-attackeď and 'non-attackeď spruce vegetation are highly complex.

Differentiating between the classes of infestation using decision tree classification only resulted in a partial, or low to moderate differentiation. The reasons for such a poor separation between the classes could be put down to the following factors: (i) The selection of classes was carried out under the use of digitally available vector data on bark beetle infestation from 1988 to 2011. By scrutinising the vector data, several infestation classes were found to be either very close to or directly next to other infestation classes. The HyMAP hyperspectral data with a pixel size of 4 m $\times$ 4 m in the border regions of two infestation classes leads to mixed information that hinders a separation of the classes. (ii) Particularly in the assignment classes from 1988 to 2008 there is a phenomenon that following the death of spruce stands, forest ecology finds its equilibrium through forest regeneration. This means that in addition to deadwood, there is also a high number of spruce and other broad-leaved saplings. This regeneration alters the spectral signature of the data, making a clear distinction between the spruce classes difficult. (iii) Apart from being influenced by vitality, the signature of the spruce vegetation is also influenced by the spruce stand age, the ground vegetation layer or regeneration. In particular, the influence of gleaming vegetation as well as spruce regeneration is in a position where it is able to superimpose and thus also falsify the typical signature of the tree stands investigated.

3.3. Scaling effects on predicting bark beetle

On 2009-08-25, hyperspectral data of the Bavarian Forest National Park "old region" was recorded using the hyperspectral sensor HyMAP at the two different ground resolutions of 4 m and 7 m. The goal of this analysis was to investigate the scaling effects from a 7 m ground resolution hyperspectral data set in terms of the predictive power of the different spruce classes. For this, we used the same model approach and sampling design as in the decision tree classification (ID3) analysis described above. Fig. 9 a and b shows the position of the mean spectral reflectance curves of HyMAP remote-sensing data with a ground resolution of 4 m and 7 m. As one can see, there is a shift in the mean spectral reflectance curves of all of the five spruce vegetation classes investigated.

In addition to a change in the mean spectral reflectance values of HyMAP data for each spruce class, for a differentiation among all five spruce classes with the decision tree (ID3) classification algorithm, only a total classification accuracy of 29.38% was achieved with a Kappa value of 0.117. The spruce class precision for all spruce classes investigated was between 25% and 37% (Table 9).

Our investigations on the influence of scaling effects on bark beetle prediction demonstrated that by increasing the spatial ground resolution from 4 m to 7 m, the predictive power deteriorated when differentiating between different vitality classes. One reason for this is that for the 4 m hyperspectral remote-sensing data with a spatial resolution of 4 m × 4 m "cleaner" pixel information was obtained for each spruce vitality stage compared to the more "mixed pixels" (many objects or features per pixel) that amalgamate a greater variety of elements from the spectral data when using a spectral ground resolution of 7 m.

4. Conclusion and outlook

The bark beetle is a very dangerous and destructive insect infesting mature spruce forests. At this moment in time we are not aware of any available method that uses remote-sensing techniques to assess or predict potential areas of bark beetle infestation in the future.

This study therefore aimed to detect spectral differences in spruce vegetation using hyperspectral remote-sensing techniques based on the vegetation's vitality classes to obtain indicators for forecasting potential bark beetle outbreaks in the Bavarian Forest National Park in Germany. Several spectral bands, vegetation indices and spectral band combinations of hyperspectral HyMAP remote-sensing data were analysed and compared in terms of their classification accuracy using the ID3 (Iterative Dichotomiser 3) decision tree classification. Furthermore, it was also tested as to which role the spatial resolution of the hyperspectral data plays in the predictive power of the forecasting model for bark beetle outbreaks.

In summary, we were able to make the following conclusions from our paper:

- The best spectral characteristics, spectral indicators and spectral derivatives in relation to vitality status and a differentiation of the spruce vitality classes were typically based on wavebands related to prominent chlorophyll absorption features in the VI over a spectral range from 450 to 890 nm. This shows that it is primarily differences in biochemical–biophysical characteristics of the photosynthetically-active pigments such as chlorophyll content that differentiates the spruce vegetation.
- Only a small amount of spectral information and derivatives were found in the SWIR1 region in the spectral range of 1400–1800 nm, which is the region that records the water content of the spruce needles.
- Differences in the spectral reflectances of the healthy, red–grey attacked vitality classes versus the "not yet attacked" vitality classes (but attacked in subsequent years) were estimated with moderate to good accuracy using hyperspectral remote-sensing data.
- Clear spectral differences with a total classification accuracy of 64.04% could be ascertained between the spruce vitality class 3 (infestation 2010; bark beetle attack in the subsequent year) and spruce class 5 (healthy).
- The vitality class C3 – green foliage but with reduced vitality (possibly the first stages of green-attack) showed a trend towards detectability and differentiation compared to the spruce class – healthy (C5) with spectral indicators and index derivatives, however with an accuracy that is regarded as insufficient in forestry practises.
- The hypothesis cannot be confirmed that the spectral differences found between spruce class 3 (infestation 2010) and spruce class 5 (healthy) were due to vegetation stress previously caused by climate change or environmental stress factors. This is substantiated by the fact that spectral differences between spruce class 4 (green and vital, but attacked 2 years later on) and spruce class 5 (healthy) could only be ascertained with a total classification accuracy of 62.34%.
- Climate-change induced changes to the biochemical–biophysical vegetation parameters in the spruce stands are temporally drawn-out and theoretically one should be able to observe them in the spectral differences of the spruce vegetation class 4 (green and vital, but infested later on in 2011) when compared to class 5 (healthy). The spectral similarity of both spruce classes 4 and 5 is however very high. The spectral differences between spruce class 3 (infestation 2010, bark beetle attack in the subsequent year) and class 5 (healthy) could possibly be a result of the first green-attack of the trees –although this is not yet visible.
- The best spectral differentiation of spruce forests was ascertained for spruce class 1 (dry 1, forest affected by bark beetle outbreaks from 1988 to 2008), spruce class 3 (infestation 2010, bark beetle attack in the subsequent year) and class 5 (healthy) with a total classification accuracy of 69.27% and a Kappa of 0.539.
- Hyperspectral data with a ground resolution of 4 m were found to contain more information relevant for estimating the vitality status of spruce vegetation compared to hyperspectral data with a 7 m ground resolution. Hence, the accuracy of the predictive model improves depending on the ground resolution of the hyperspectral remote-sensing data.

Further investigations using hyperspectral remote-sensing data are necessary to substantiate these results. Given the importance of forests as carbon sinks in the face of climate change and the widespread destruction of forests caused by the bark beetle, the authors believe that this topic warrants further investigation. We therefore aim to conduct future research that will explore the relationship between the spectral response of hyperspectral signals and spruce leaf biochemical–biophysical components in systematic climate-stress scenarios, based on the experimental approach of Lausch et al. (2012). Consequently, we expect to be able to make tightly-focussed statements about the spectral characteristics on the different vitality classes of spruce vegetation by excluding disturbance effects such as different image textures, BRDF effects, sensor-characteristics as well as illumination effects on spruce vegetation. Another interesting aspect of future research will be to incorporate new flight campaigns using the hyperspectral sensor AISA (Lausch et al. 2013a,b) with a higher ground resolution of 2 m, from which we expect to be able to make more accurate classifications and contribute to closing the gap that still remains in the research on green-attack detection methods.

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